Malaria Cell Detection: Final Study Guide

Everything needed to explain the whole project at the final submission, in the order of the course brief. Read it once with the deck open (about 90 minutes). Before the defence, go over the numbers card (section 11) and the question drill (section 12).

Team	J Amathziah (230139) · C Murali Madhav (230115) · Ravi Yadav (230131) · Pugazhendhi J (230066)
Topic	Malaria cell detection on the NIH Malaria dataset (one of the brief's eight topics)
Base paper	G. Marques, A. Ferreras, I. de la Torre-Díez, *An ensemble-based approach for automated medical diagnosis of malaria using EfficientNet*, Multimedia Tools and Applications 81 (2022)
Our change	YUV histogram-equalisation stain normalisation (H2), tested on patients the model has never seen (H1: a patient-level split)
Deliverables	IEEE paper `Malaria_IEEE_Paper.pdf` in the repository's top folder (8 pages) · deck `slides/Malaria_Phase3_Final.pptx` (27 slides) · code, results and all 40 trained models on GitHub

Contents

0. The brief, and where we answered each part
1. The whole project in two minutes
2. Step 1 · Literature survey
3. Step 2 · Base paper and reproducibility checklist
4. Step 3 · Reproducing the paper
5. Step 4 · Hypotheses and predictions, fixed before coding
6. Step 5 · The experiment: design, statistics, results
7. Why our predictions were wrong
8. Step 6 · Failure analysis
9. Limitations, and how to talk about them
10. The paper and the code, section by section
11. Numbers card
12. Question drill: 32 likely questions
13. Glossary of the Phase 3 terms

Deeper background on the dataset, EfficientNet-B0, the training recipe and every Phase 2 metric is in the Project Guide. This guide summarises it where needed.

The brief, and where we answered each part

The brief asks us to pick a healthcare vision topic, reproduce a paper, then propose and test **one change of our own**, and write it up like a conference paper. Marks come from **predictions, experiments and defence**.

Step (milestone)	The brief asks	What we did	Where to point
1 · Literature survey (M1)	Read 10 to 20 recent papers; note methods, datasets, metrics	11 malaria studies (2016 to 2024), plus 7 methods papers. Found three gaps	Slide 6 · paper §II
2 · Pick a base paper (M1)	One paper with public code and data; fill the reproducibility checklist	Marques et al. 2022: executed notebook public, NIH data public	Slide 7 · `docs/REPRODUCIBILITY_CHECKLIST.md`
3 · Reproduce (M2)	Re-run on the same data and land near the reported metric	Code-exact, all 10 folds: ensemble 97.62 % vs 98.29 %; fold models 97.55 % vs 97.56 %	Slides 8 to 11 · paper §IV-A, §V-A
4 · Hypothesis + Introduction (M3)	One change, its predicted effect and why, submitted before coding	H1 and H2 in the Phase 2 deck; analysis plan committed on 2 Nov at 03:24, before any A1 to A3 model was trained	Slide 13 · paper §I · `docs/phase3_analysis_plan.md`
5 · Implement and test (M4)	Make the change, run ablations, measure what moved	A 2 × 2 ablation (A0 to A3), 30 new models	Slides 14 to 20 · paper §IV-B, §V-B, §V-C
6 · Analyse and write (M5)	Failure analysis and metrics; IEEE paper; defend it	Errors by patient, false alarms, label-error candidates, blind triage; recall-first threshold; patient-level rule; 8-page IEEE paper	Slides 21 to 26 · paper §V-D to §VII
Rule 1	Predict before you experiment; being wrong is fine	Both predictions turned out wrong, and we can say why	Slide 21 · section 7
Rule 2	No clinical claims; report recall and AUC; include limitations	Recall, specificity, MCC and ROC-AUC for every experiment; a limitations slide and section	Slides 17, 23, 25 · paper Tables III to V, §VII

The "one change" question. Each experiment changes exactly one thing. Our change to the method is **H2, YUV stain normalisation**. H1 does not change the model: it changes how we *test* it (split by patient), so that H2 is measured on patients the model has never seen. A3 is the control.

The whole project in two minutes

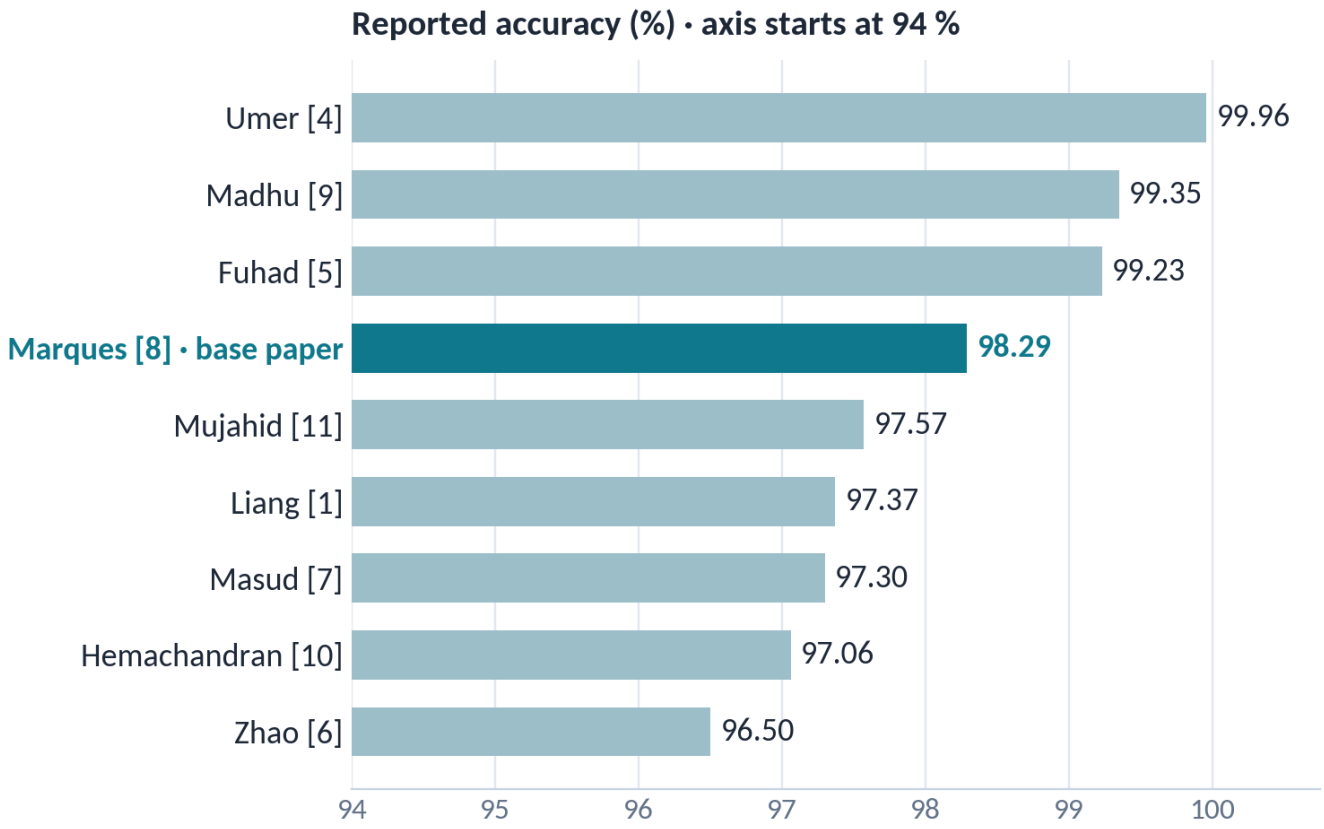
1. **The problem.** Malaria is diagnosed by a technician counting Giemsa-stained blood cells under a microscope. This is slow, needs scarce experts, and varies between observers. CNNs report 96 to 99.9 % accuracy on the public NIH cell images.
2. **The catch.** Most studies split the *images* at random, so cells of the same patient sit in both the training and the test set (**patient leakage**). Only Rajaraman et al. test on unseen patients.
3. **Reproduction.** We rebuilt the EfficientNet-B0 ten-fold ensemble of Marques et al. (2022) from the authors' own executed notebook.
    - Ensemble accuracy: 97.62 % against their 98.29 %, with every metric within 1 pp.
    - Their text swaps precision and recall, and their code scored augmented test images.
    - Our training runs about 25× faster.
4. **Hypotheses, fixed in advance.**
    - H1: a patient split lowers the scores.
    - H2: YUV stain normalisation helps on unseen patients.
5. **Results.**
    - H1 is **not supported**. The ensemble scores **98.45 %** on 40 unseen patients. Measured on identical cells, leakage is real but small: **0.85 pp of MCC**.
    - H2 is **rejected**. MCC changes by **–0.26 pp**, which is within ±1 pp of zero.
6. **Failure analysis.**
    - Errors cluster in a few patients.
    - Almost all false alarms fall on infected patients' smears, which fits possible label noise.
    - A per-patient rule finds 26 of 28 infected and 12 of 12 healthy unseen patients. This is exploratory and uses curated cells.

Say it in one breath. "We rebuilt a published 98 % malaria classifier from its authors' own code and landed within one point. We predicted that its accuracy would fall on patients it had never seen, and that stain normalisation would help there. Both predictions were wrong. The accuracy survives (98.45 %), leakage is real but small (under one point of MCC), and the normalisation does nothing measurable. Every test was fixed before we ran it, so these results are as trustworthy as a positive one."

Step 1 · Literature survey

We reviewed **11 studies** on the NIH cell images, from Liang et al. (2016) to Mujahid et al. (2024), plus the methods papers we rely on:

- EfficientNet and noisy-student training, for the model;
- Nadeau-Bengio, Holm and Lakens, for the statistics;
- Northcutt, for label errors;
- Tellez, for stain augmentation.



Headline accuracy as each study reports it. Protocols differ, so these numbers are not directly comparable.

Study	Approach	Reported result
Liang et al. 2016	16-layer CNN from scratch (transferred AlexNet features: 91.99 %)	97.37 %
Rajaraman et al. 2018	Released the dataset. Pre-trained CNNs as feature extractors; also evaluated per patient	98.6 % per cell, 95.9 % per patient
Rajaraman et al. 2019	Ensemble of VGG-19 and SqueezeNet; patient-level evaluation	(ensemble study)
Umer et al. 2020	YUV conversion and luma histogram equalisation before a stacked CNN	≈ 99.9 %
Fuhad et al. 2020	Light CNN on a phone, after setting aside 5.1 % of the cells as mislabelled or suspicious	99.23 %

Study	Approach	Reported result
Zhao et al. 2020	Extends detection to a second species	96.5 %
Masud et al. 2020	409 K-parameter CNN in a mobile app	97.3 %
Marques et al. 2022 (base)	**EfficientNet-B0, ten-fold ensemble**	**98.29 %, ROC-AUC 99.76 %**
Madhu et al. 2023	Inception-based capsule network	99.35 %
Hemachandran et al. 2023	Comparison of backbones	97.06 %
Mujahid et al. 2024	Comparison of backbones	97.57 %

The three gaps we found.

1. **Patient leakage.** Most studies split images at random. Only Rajaraman et al. (2018, 2019) evaluate on unseen patients, and their scores drop there.
2. **The numbers are not comparable.** The studies use 5-fold, 10-fold or 80:20 protocols, on original or cleaned labels.
3. **The work is rarely reproducible.** Few studies publish their full training code. Marques et al. published their executed notebook.

Say it like this. “Eleven studies report 96.5 to 99.9 per cent, but they can’t be compared directly, and only Rajaraman’s group tests on patients the model hasn’t seen. That gap is our project.”

Step 2 · Base paper and reproducibility checklist

Why Marques et al. (2022)?

- It is a strong result: 98.29 % accuracy and ROC-AUC 99.76 %.
- It uses a standard architecture.
- Its authors published their executed notebook as a supplementary file, so we could rebuild the recipe exactly.
- The data are public.

The reproducibility checklist (full version in `docs/REPRODUCIBILITY_CHECKLIST.md`, slide 7):

Item	In the paper?	What we found, and what we did
Dataset public	✓ NIH Malaria, 27,558 cells	Downloaded automatically; patient IDs read from the file names and checked against NIH's own mapping
Code public	✓ executed notebook	We follow the code, not the text, wherever they differ
Library versions	Partly, in the notebook	TensorFlow 2.3 then; we use 2.15, the last Keras 2 release, with the same `efficientnet` 1.1.1 and `albumentations` 0.5.2
Pre-trained weights	Text: ImageNet	Code: noisy-student. We follow the code
Hyperparameters	The text differs from the code in 8 places	Listed on slide 8; we follow the code
Data split	80/20 then 10 folds, **unseeded**	Cannot be recovered. We sort and seed the split and store its fingerprint with every fold
Evaluation protocol	Not stated	The code scored **randomly augmented** test images. We report both protocols
Metrics	Accuracy, precision, recall, F1, AUC	The text **swaps precision and recall**; Table 6's "accuracy" is (precision + recall) / 2
Model size	Table 2: 4,223,934 parameters	Matched exactly, and a test checks it
Hardware and time	GTX 1050, 4 h 46 min per fold	RTX 4090, about 11 min per fold
Patient-level evaluation	✗ random image split	This became our H1
Trained models released	✗	We release all 40 of ours (`v1.0-a0-10fold`, `v2.0-ablation`)

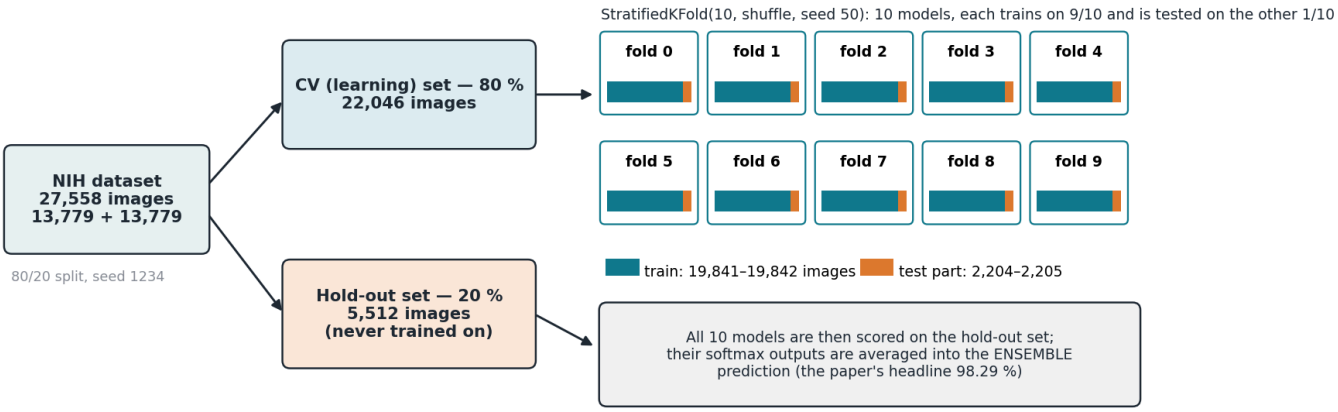
The code differs from the text (slide 8):

Setting	Paper's text / our first guess	Authors' code (what we follow)
Weights	ImageNet	noisy-student, qubvel `efficientnet` 1.1.1
Head	GAP → dropout → 1 sigmoid	GAP → 128 → 64 → 32 (dropout 0.3) → 2 softmax
Loss	binary cross-entropy	categorical cross-entropy, label smoothing 0.1
Schedule	≤ 15 epochs, early stopping	33 epochs, LR halved on plateau, best checkpoint
Batch size	128	16 (1,240 steps per epoch)
Hold-out	10 %	20 % = 5,512 images, never trained on
Augmentation	flips, 90° rotations	heavy `albumentations`: noise, blur, distortions, CLAHE…
Evaluation	clean images	augmented images (one generator for all)

Step 3 · Reproducing the paper

The data

- 27,558 single-cell crops: 13,779 parasitized and 13,779 uninfected.
- 200 patients (150 infected with *P. falciparum*, 50 healthy), photographed with a smartphone on a microscope at Chittagong Medical College Hospital, Bangladesh.
- **The patient is in the file name.** `C33P1thinF_IMG_..._cell_179.png` belongs to patient C33. All 27,558 cells map to NIH's patient list.
- **Both classes come from the same people.** 10,364 of the 13,779 uninfected cells (75 %) come from *infected* patients' smears, so almost every patient has cells in both classes.
- Patients differ widely: from 65 to 702 cells each.

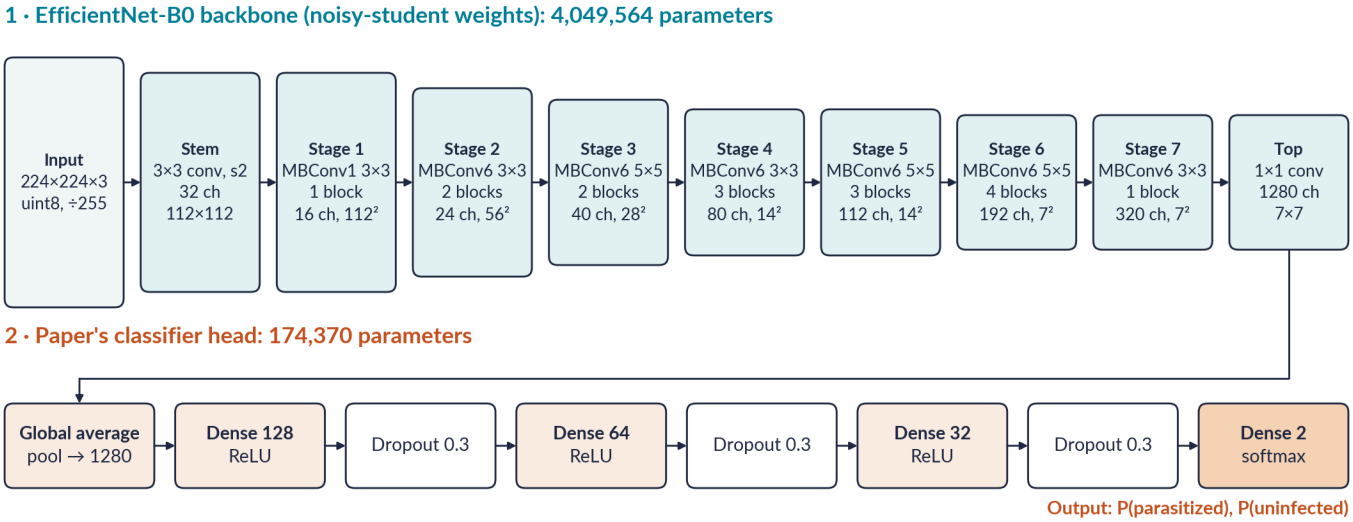


The paper's split: 80 % learning set in ten folds, 20 % hold-out set that no model trains on.

The paper's split.

- The data are split 80/20: 22,046 learning cells and 5,512 hold-out cells that no model ever trains on.
- The learning set is cut into 10 stratified folds. Each fold's model trains on 9 parts and is checked on the 10th.
- The 10 models are then averaged on the hold-out set.

The model



Box height shows the resolution (224 → 112 → 56 → 28 → 14 → 7 pixels) as the channels grow 3 → 32 → … → 320 → 1280.
16 MBConv blocks in 7 stages · 81 convolution layers (65 normal + 16 depthwise) · 49 batch-norm layers
230 Keras layers in the backbone · 4 dense layers in the head

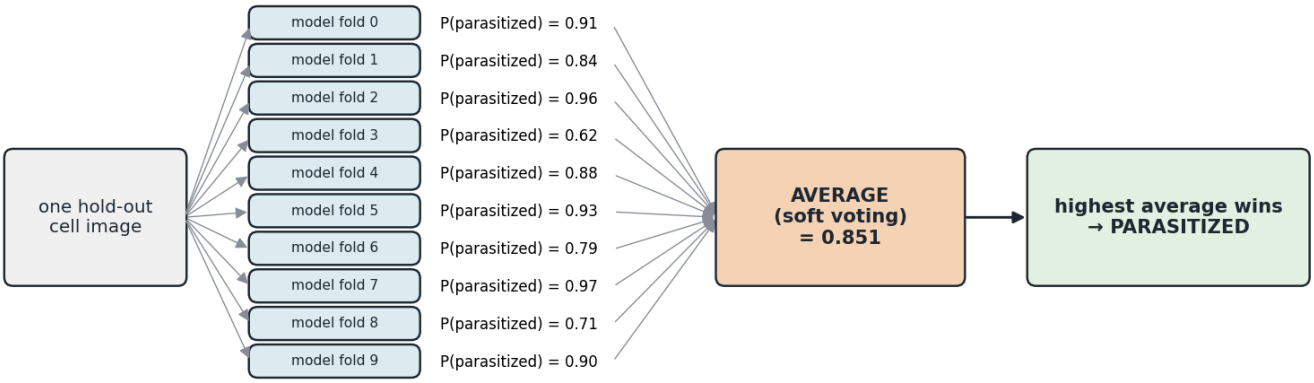
Total: 4,223,934 parameters (4,181,918 trainable + 42,016 non-trainable batch-norm statistics), exactly the paper's Table 2.

EfficientNet-B0 with the paper's head: 4,223,934 parameters, exactly the paper's Table 2.

- **Backbone.** A stem convolution and 16 MBConv blocks in seven stages turn a 224 × 224 × 3 cell into 7 × 7 × 1,280 feature maps.
- **MBConv block:** expand (1 × 1 convolution) → depthwise convolution → squeeze-and-excitation → project (1 × 1) → skip connection.
- **Head.** Global average pooling gives 1,280 numbers. Dense layers of 128, 64 and 32 units follow (ReLU, dropout 0.3), then a two-way softmax: P(parasitized) and P(uninfected).
- **Size.** 4,223,934 parameters, 4,181,918 of them trainable.

The training recipe (from the authors' code)

- Categorical cross-entropy with **label smoothing 0.1**: targets of 0.95/0.05 instead of 1/0.
- Adam, learning rate 10^{-4}. The rate is halved when the validation loss stalls for 6 epochs, down to 10^{-6}.
- **33 epochs** of **batch 16** (1,240 steps per epoch); the best-validation-loss weights are restored at the end.
- Heavy random augmentation: flips, rotations, noise, blur, distortions, CLAHE, sharpening, embossing, and contrast and brightness changes.



Ten models that each make different small mistakes, averaged, make fewer mistakes than a typical single one (illustrative numbers).

The ensemble: each of the ten fold models gives P(parasitized) for a cell, and the ten probabilities are averaged (soft voting).

Result: within 1 pp of the paper on every metric

Metric (hold-out, 5,512 cells)	Paper	Ours (paper protocol)	Δ (pp)
Mean fold accuracy	97.56 %	97.55 %	−0.01
Ensemble accuracy	**98.29 %**	**97.62 %**	**−0.67**
Recall (sensitivity)	97.74 %*	97.07 %	−0.67
Precision	98.82 %*	98.11 %	−0.71
Specificity	98.84 %	98.17 %	−0.67
F1	98.28 %	97.59 %	−0.69
ROC-AUC	99.76 %	99.73 %	−0.03

*Corrected values. The paper's text swaps precision and recall.

Our ensemble missed 80 of 2,730 infected cells and flagged 51 of 2,782 healthy ones (the paper: 62 and 32).

Why the ensemble is 0.67 pp short.

- The paper's random split had no seed and cannot be recreated, and we use a newer TensorFlow.
- Our ensemble gains 0.25 pp over its average member, where the paper's gained 0.60 pp.
- The gap is about three standard errors on 5,512 cells. Single models match the paper to 0.01 pp.

What reading the authors' code revealed:

1. The paper's text swaps the ensemble's precision and recall. The true recall is 97.74 %, not 98.82 %.
2. Every reported prediction was made on a randomly augmented image. Scoring clean images changes our numbers by at most 0.23 pp, so this doesn't explain their result.
3. Table 6's "accuracy" column is (precision + recall) / 2 of each printed report.

Faster, with the same maths:

- every image is decoded once into a shared `uint8` cache;
- the augmentation runs in 11 parallel workers, each batch seeded by (fold, epoch, batch);
- batches go to the GPU as bytes and are scaled inside the model;
- the training step is XLA-compiled, which takes it from 21 to 12.5 ms;
- 10 cloud GPUs run the 10 folds in parallel.

The result is about **11 min per fold on an RTX 4090, against 4 h 46 min** on the paper's GTX 1050. Arithmetic stays float32.

How we know the rebuild is faithful. 32 automated tests check it against the paper's printed outputs:

- the parameter counts of Table 2;
- the split and fold sizes;
- the augmentation pipeline, verbatim;
- the image loading, pixel for pixel;
- the loss, optimiser and callbacks;
- the classification reports.

Step 4 · Hypotheses and predictions, fixed before coding

	H1 · validation	H2 · preprocessing (our change)
Change	Split by patient, not by image; model unchanged	RGB → YUV → equalise the brightness (Y) histogram → RGB (Umer et al. 2020), on the patient split
We expected	Accuracy and MCC fall below the image-split baseline; the fold-to-fold spread widens	Higher recall and MCC than A1 on unseen patients; little change on the random split
Because	A patient's cells share stain, lighting and blood, so a random split lets the model recognise the patient rather than the parasite	Equalising brightness removes slide-specific cues, so the network has to rely on the parasite
Falsified if	Patient and image splits score the same	The gain over A1 is smaller than A1's fold-to-fold SD

The formal decision rules (`docs/phase3_analysis_plan.md`):

- **H1 is supported** if A1's ensemble MCC is lower than A0's, with the 95 % CI of the difference excluding 0 (Holm $p < 0.05$).
- **H2 is rejected** if A2's mean fold-MCC gain over A1 is smaller than A1's fold-MCC SD.
- **The control** counts as "little change" if |A3 – A0| in mean fold MCC is below A0's fold-MCC SD.
- **Primary metric:** MCC. **Primary protocol:** clean (un-augmented) evaluation.

The timeline is our proof that we predicted before experimenting:

When	What	Evidence
Phase 2 (October)	H1, H2 and their falsification rules presented	Phase 2 deck, slides 28 to 29
2 Nov, 03:24	Analysis plan and notebook committed, **before any A1 to A3 model was trained**	commit `0c086ee`, `docs/phase3_analysis_plan.md`
2 Nov, morning	30 models trained on RunPod (about 1 hour)	`results/A1_*`, `A2_*`, `A3_*`; release `v2.0-ablation`
2 Nov, 17:32	Four exploratory analyses fixed in a dated addendum, **before running them**	commit `8e17fd6`, `docs/phase3b_addendum.md`
2 Nov, evening	Phase 3b analyses run on the saved predictions	`results/phase3b/`

The paper's Introduction (§I) follows the brief's M3:

- the clinical problem;
- two questions: can the published numbers be reproduced, and do they hold on unseen patients?
- five contributions:
  1. a code-exact reproduction;
  2. three corrections to the paper's reporting;

3. a pre-registered one-variable ablation;
4. a patient-level statistical analysis;
5. exploratory analyses fixed in an addendum.

Step 5 · The experiment: design, statistics, results

The 2 × 2 design: each comparison changes one thing

ID	Split	Preprocessing	Question it answers
A0	image (the paper's)	none	the baseline (Phase 2)
A1	patient	none	**H1:** A1 vs A0, does leakage inflate the score?
A2	patient	YUV + hist.-eq.	**H2:** A2 vs A1, does normalisation help on unseen patients?
A3	image (the paper's)	YUV + hist.-eq.	**Control:** A3 vs A0, does YUV matter when patients are shared?

Held fixed in every experiment:

- the model and its weights;
- the loss and optimiser;
- 33 epochs and batch 16;
- the augmentation;
- the per-fold seeds;
- the 10-model ensemble.

A1 and A2 use the identical split, and A3 reuses A0's split cell for cell, so both of those comparisons are **paired** (the same cells in both arms).

Switch 1: the patient split (`src/marques2022_exact.py`)

```python
def patient_id(file):
    """The patient of an NIH cell image: the leading C-number of its file name
        ("C37BP2_thinF_..." -> "C37")."""
    m = re.match(r"^(C\d+)", os.path.basename(str(file)))
    return m.group(1)

def patient_split(df, seed=1234, outer_splits=5):
    y, groups = _binary_labels(df), df["file"].map(patient_id).values
    outer = StratifiedGroupKFold(outer_splits, shuffle=True, random_state=seed)
    candidates = list(outer.split(df, y, groups))
    # take the candidate closest to 20 % of the cells and 50 % parasitized (a rule on labels
        only)
    scores = [abs(len(hold) / len(df) - 0.2) + abs(y[hold].mean() - 0.5) for _, hold in
        candidates]
    learn_pos, hold_pos = candidates[int(np.argmin(scores))]
    return df.iloc[learn_pos], df.iloc[hold_pos]
```

How it works.

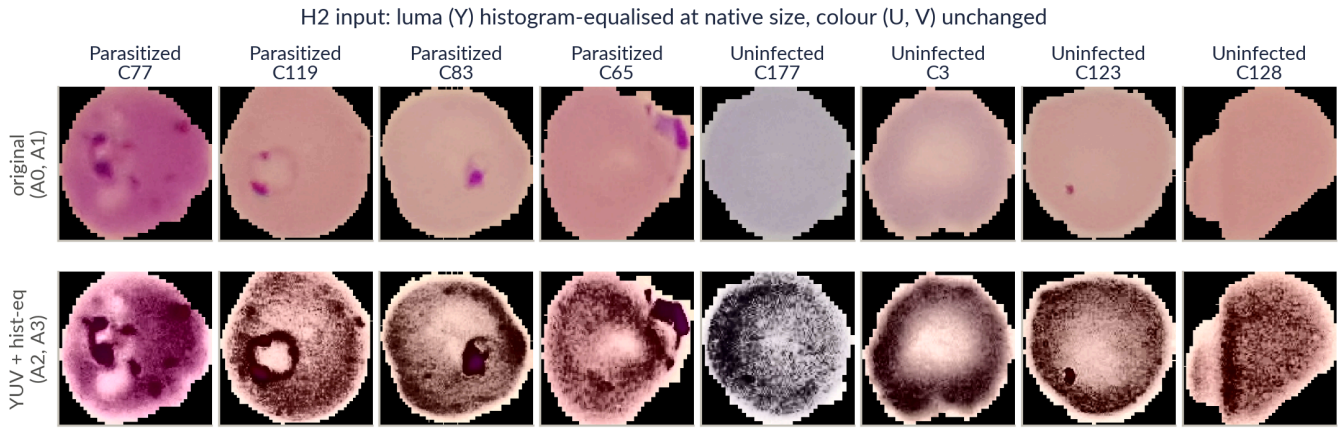
- `StratifiedGroupKFold` keeps every patient's cells together (*group*) while balancing the classes (*stratified*).
- The rule picks one of five candidate hold-out sets using labels only, never model scores.

What it produces.

- A hold-out set of **5,550 cells from 40 patients** (28 infected, 12 healthy), 50.7 % parasitized.
- The other **160 patients** form 10 folds of 15 to 17 patients each.
- The split is committed to the repository, and every training run checks its fingerprint (`ff94f8c3b200e708`).
- `check_no_shared_patients` raises an error if any patient appears on both sides.

Switch 2: YUV stain normalisation

```
def yuv_equalize(bgr):
    yuv = cv2.cvtColor(bgr, cv2.COLOR_BGR2YUV)
    yuv[..., 0] = cv2.equalizeHist(yuv[..., 0])     # equalise brightness (Y) only; colour (U,
        V) untouched
    return cv2.cvtColor(yuv, cv2.COLOR_YUV2BGR)
```



Top: original cells from eight patients. Bottom: after YUV conversion and luma histogram equalisation. The narrow brightness range of each cell is stretched over the full range, which also amplifies texture.

How the step is applied.

- It runs at each image's native size, *before* the unchanged resize and augmentation.
- The black background stays black.
- We left out Umer's 120 × 120 resize and their edge kernel, which the paper does not specify.

Key point for the defence: the step changes **brightness only**. It does not touch stain *colour*.

How we decide: the statistics in plain words

1. MCC, the primary metric.

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Worked example: the A1 ensemble on 5,550 cells.

- TP = 2,760 (parasites found), FN = 55 (missed), FP = 31 (false alarms), TN = 2,704.
- Accuracy = (2,760 + 2,704) / 5,550 = 98.45 %.
- Recall = 2,760 / 2,815 = 98.05 %.
- Specificity = 2,704 / 2,735 = 98.87 %.
- MCC = 0.9690, which we write as **96.90** (× 100).

Why MCC?

- It uses all four boxes of the confusion matrix and is high only when *both* classes are predicted well.
- Unlike accuracy, it is not flattered when one class dominates. The class mix varies between patient folds.

2. The patient bootstrap, for honest confidence intervals.

```
def _patient_weights(groups, n_boot, rng):
    """Cell weights for each bootstrap resample: how often the cell's patient was drawn."""
    _, inv = np.unique(np.asarray(groups), return_inverse=True)
    n_groups = inv.max() + 1
    for _ in range(n_boot):
        yield np.bincount(rng.integers(0, n_groups, n_groups), minlength=n_groups)
        [inv].astype(float)
```

What it does. We draw 40 patients *with replacement*, 10,000 times. Each time we recompute the metric, and the middle 95 % of those values is the 95 % CI.

Why whole patients? A patient's cells share stain, lighting and blood, so they are not independent. With cells resampled one by one, the interval would be too narrow. Patient clustering inflates the variance **2.3×** for accuracy and **2.6×** for MCC (the *design effect*).

3. Paired versus unpaired comparisons.

- **Paired:** A2 vs A1, and A3 vs A0, score the same cells. Both arms see the same resample, so noise cancels. This design can detect an effect of about **0.77 pp** of MCC.
- **Unpaired:** A1 vs A0 have different hold-out sets, so each arm resamples its own patients. This could detect only about **2.2 pp**. That is why we later measured leakage on identical cells.

4. The Nadeau-Bengio corrected t-test, for fold scores.

```
def corrected_resampled_ttest(diffs, n_train, n_test):
    """Nadeau & Bengio (2003): t = mean(d) / sqrt((1/k + n_test/n_train) * var(d)), df = k -
        1."""
    d = np.asarray(diffs, float)
    k, var = len(d), d.var(ddof=1)
    t = d.mean() / np.sqrt((1 / k + n_test / n_train) * var)
    return float(t), float(2 * sps.t.sf(abs(t), df=k - 1))
```

Why it is needed. Any two of the ten fold models share 8 of their 9 training parts, about 89 % of their data. So their scores are correlated, and a plain t-test is far too confident. The extra term `n_test / n_train` (= 1/9) widens the interval to account for that.

Our correction. Our plan used a plain Welch test for the unpaired H1 fold comparison; it gave Holm p = 0.023. We corrected it to p = 0.14 and report both values.

5. The Holm correction, for three contrasts at once.

- Sort the three p-values. Multiply the smallest by 3, the next by 2 and the largest by 1, and never let a value fall below the one before it.
- **Example:** 0.01, 0.03 and 0.04 become 0.03, 0.06 and 0.06.
- **Why:** testing three things at once raises the chance of a false "significant".

6. Equivalence (TOST) and the minimum detectable effect.

- **“Not significant” does not mean “no effect”.** It can also mean the test was too weak to see one.
- **TOST** asks something stronger: is the 90 % CI entirely inside a margin we call negligible, ±1 pp of MCC? If so, the effect is practically zero.
- **MDE (minimum detectable effect)** is the smallest true effect a design detects 80 % of the time.

```
def tost_from_boot(boot, margin=0.01):
    lo, hi = np.percentile(boot, [5, 95])                    # 90 % interval of the bootstrapped
        difference
    return dict(lo90=lo, hi90=hi, equivalent=bool(-margin < lo and hi < margin))
```

Results: every metric, all four experiments (clean evaluation)

	A0	A1	A2	A3
Split / preprocessing	image / none	patient / none	patient / YUV	image / YUV
Hold-out set	5,512 cells	5,550 cells, 40 unseen patients	same as A1	same as A0
Ensemble accuracy	97.64 %	**98.45 %**	98.32 %	97.64 %
Recall (sensitivity)	97.18 %	98.05 %	98.22 %	97.14 %
Specificity	98.09 %	98.87 %	98.43 %	98.13 %
MCC × 100	95.29	96.90	96.65	95.29
ROC-AUC	99.68 %	99.80 %	99.76 %	99.66 %
Mean fold MCC ± SD	95.57 ± 0.60	93.92 ± 1.51	93.32 ± 1.66	95.20 ± 0.37
Mean fold accuracy ± SD	97.78 ± 0.30	97.24 ± 0.66	96.96 ± 0.75	97.60 ± 0.19
Missed / false alarms	77 / 53	55 / 31	50 / 43	78 / 52

A0 appears here under the clean protocol (97.64 %). Under the paper’s augmented protocol it scores 97.62 %, the number in section 4.

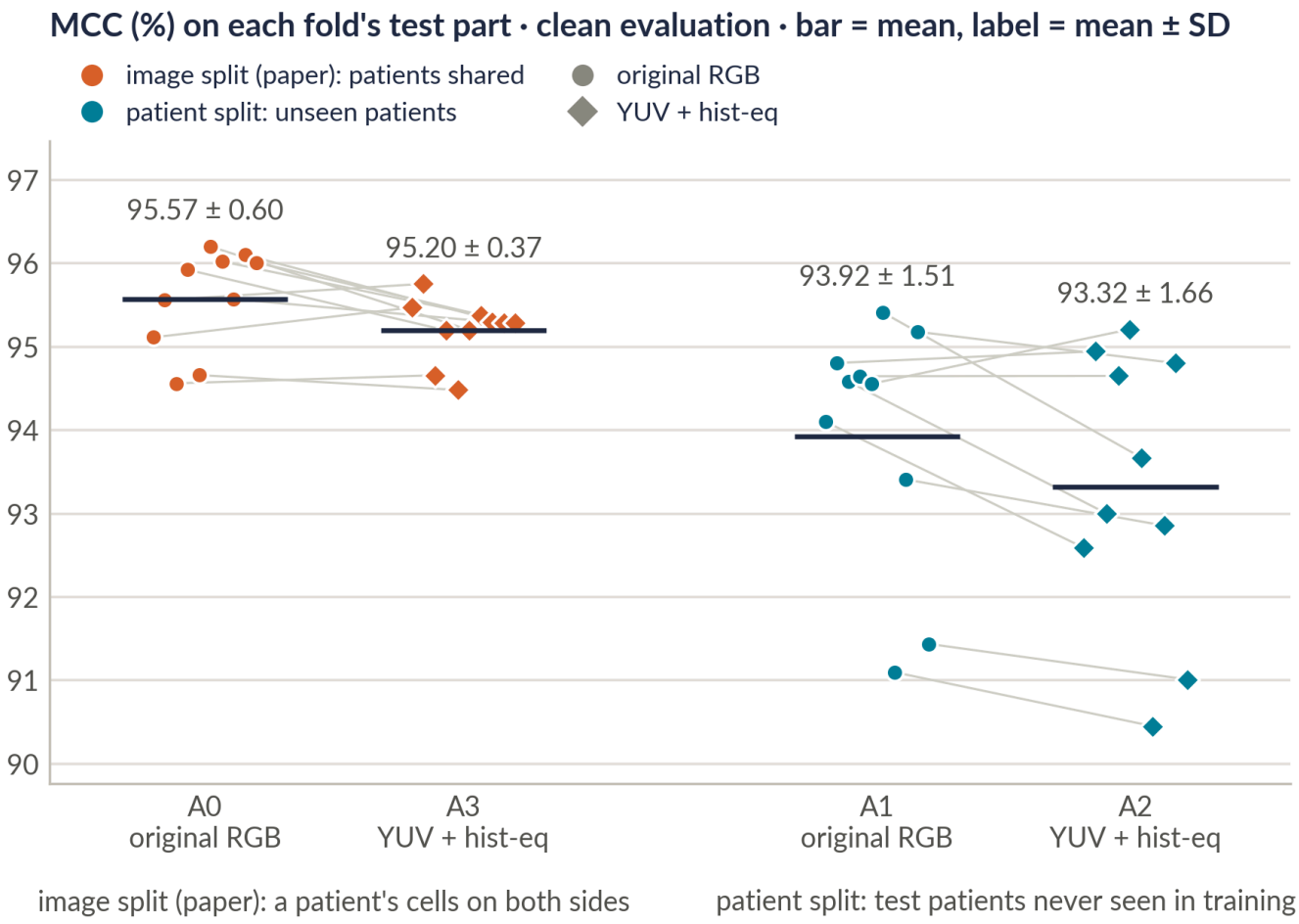
H1: does the patient split lower the scores?

We measured H1 three ways.

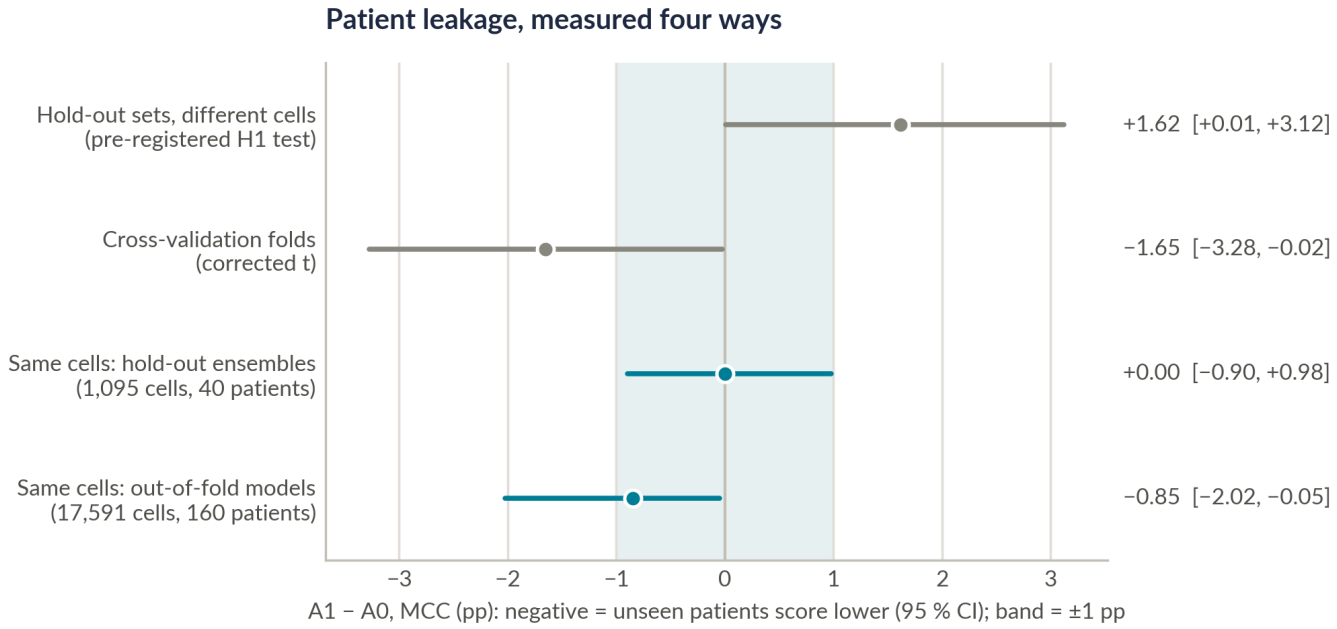
1. **The pre-registered test.** Ensembles on their own hold-out sets, which are different cells.
   - A1’s MCC is *higher*: **+1.62 pp** (95 % CI +0.01 to +3.12, Holm p = 0.15).
   - Verdict: **H1 is not supported.**
2. **Cross-validation over 160 patients.**
   - Fold MCC: **−1.65 pp** (95 % CI −3.28 to −0.02, Holm p = 0.14).
   - Fold accuracy: 97.78 → 97.24 %.
   - The fold-to-fold spread widened **2.5×** (SD 0.60 → 1.51), as H1 predicted.
   - Validation also stops improving earlier: the median best epoch is 17 instead of 28.5.
   - This is a fragile trend, not a significant effect.

3. **Identical cells (Phase 3b), the cleanest test.**
    - **17,591 cross-validation cells of 160 patients:** models that never saw the patient score **0.85 pp** lower in MCC (95 % CI 0.05 to 2.02), or 0.43 pp in accuracy.
    - **1,095 hold-out cells of the 40 patients:** the difference is **0.00** (–0.90 to +0.98). The confusion matrices are identical, and 19 of each ensemble's 22 errors fall on the same cells.

Why is the unseen-patient score higher? The 40-patient draw is easy. The same A1 fold models average 98.22 % on it but 97.24 % on the unseen patients of cross-validation. That difference of about 1 pp is larger than the leakage effect itself.



MCC of every fold model on its cross-validation test part. Orange: image split. Teal: patient split, which scores lower and varies more. YUV changes little.



A1 − A0 in MCC measured four ways (95 % CIs; shaded band: ±1 pp). The two teal rows score identical cells.

Conclusion: patient leakage is real on this dataset, but small (under about 2 pp of MCC, half a point of accuracy). A0's models also trained on more patients, so these numbers, if anything, overstate it.

H2: does YUV normalisation help on unseen patients?

The pre-registered rule.

- A2 vs A1 mean fold MCC: **−0.60 pp**, smaller than A1's fold SD of 1.51 pp, so **H2 is rejected**.
- The corrected t-test agrees (p = 0.13).

The ensembles on the 40 unseen patients:

Metric	Change (95 % CI)
MCC	**−0.26 pp** (−0.85 to +0.23)
Recall	+0.18 pp (not significant)
Specificity	−0.44 pp (not significant after Holm)
ROC-AUC	−0.04 pp

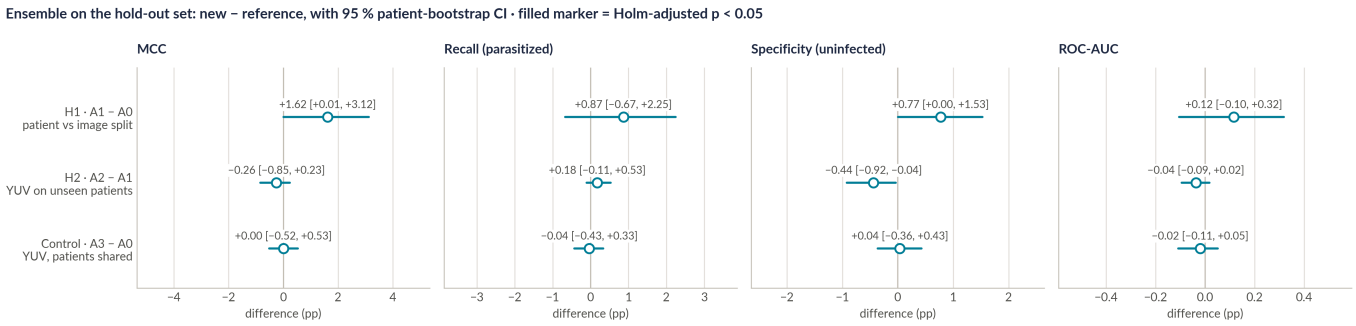
Equivalence.

- **Hold-out:** the 90 % CI is −0.73 to +0.16 pp, inside ±1 pp (TOST p = 0.008), so the effect is **practically zero**.
- **Fold level:** −1.27 to +0.06, which is too wide to show equivalence.

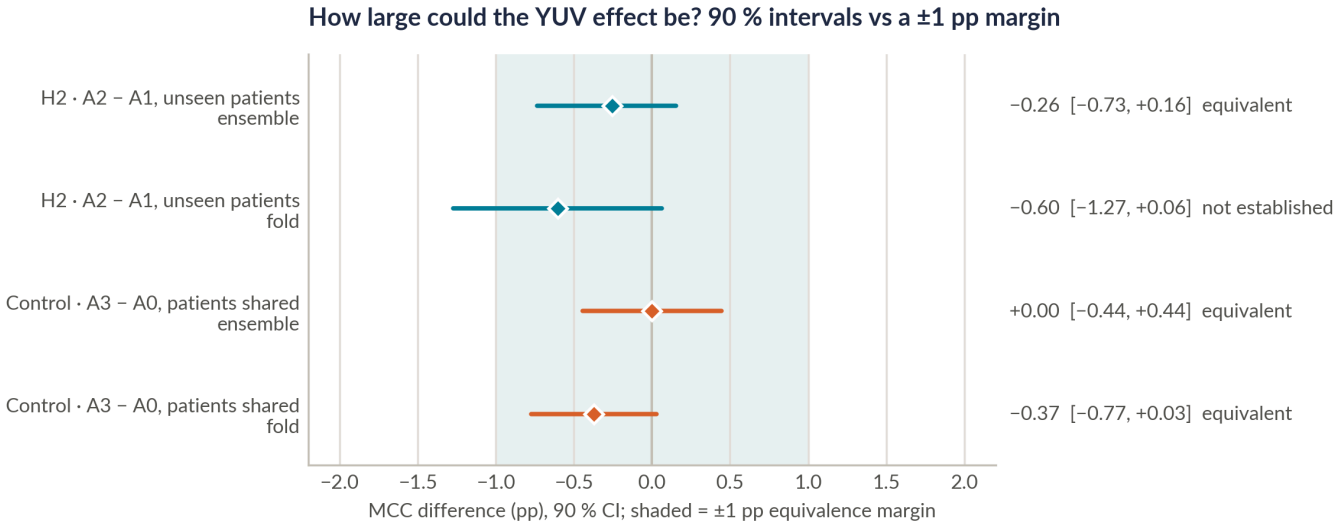
The control, A3 vs A0.

- Ensemble MCC +0.00 pp (95 % CI −0.52 to +0.53).
- Equivalent within ±1 pp at both levels.

The paper's augmented protocol gives the same three verdicts.



The three contrasts on the ensembles' hold-out sets, with 95 % patient-bootstrap intervals. None survives the Holm correction.



Equivalence: 90 % intervals of the two YUV contrasts against the ±1 pp margin.

"Not supported" and "rejected" are different verdicts. Say them the same way every time.

- **H1 had a rule for *support*.** Supporting it needed a significant drop on unseen patients, and that didn't happen. This is not proof of zero leakage: the identical-cell test finds a small amount.
- **H2 had a rule for *rejection*.** Its gain had to be smaller than the fold spread, and it was (–0.60 vs 1.51 pp).

Why our predictions were wrong

The brief says being wrong is fine and good reasoning is what counts. This section is the reasoning (slide 21).

	We predicted	What happened	Most likely why (our experiments don't separate these)
H1	A clear drop on unseen patients and a wider spread	The spread widened 2.5× ✓. The drop is small: 0.85 pp MCC on identical cells, and none on the 40-patient hold-out	(1) The decisive cue, purple-stained parasite chromatin, looks alike in every patient. (2) The paper's heavy colour, contrast and blur augmentation already stops the network from relying on a patient's staining. (3) End-to-end fine-tuning learns the parasite better than the frozen features Rajaraman et al. used, who saw 98.6 → 95.9 %
H2	Higher recall and MCC on unseen patients	MCC −0.26 pp, within ±1 pp; recall +0.18 pp; specificity −0.44 pp	(1) YUV equalisation changes **brightness only**, not stain colour. (2) The augmentation already randomises brightness and contrast, including CLAHE, so nothing is left to fix. (3) Umer et al.'s 99.96 % vs 49.61 % compared against a model that didn't train, not a strong baseline. (4) All images come from one lab, so there is little stain variation to remove. (5) Stretching each cell's narrow brightness range amplifies texture and noise, which fits the slightly lower specificity

What we learned.

- **Measure a gain against a strong baseline.** A preprocessing gain reported against a weak model may vanish against a strong one.
- **Use leakage-proof evaluation for small differences.** On a single random hold-out set, sub-1-pp differences between methods can't be resolved, because one draw of 40 patients moved accuracy by 1 pp. Cross-validate over patients, report patient-level intervals and pair the comparisons.
- **Test stain robustness the right way.** That means colour *augmentation* rather than normalisation (Tellez et al. 2019), tested on another laboratory's smears (MP-IDB). This is our next experiment.

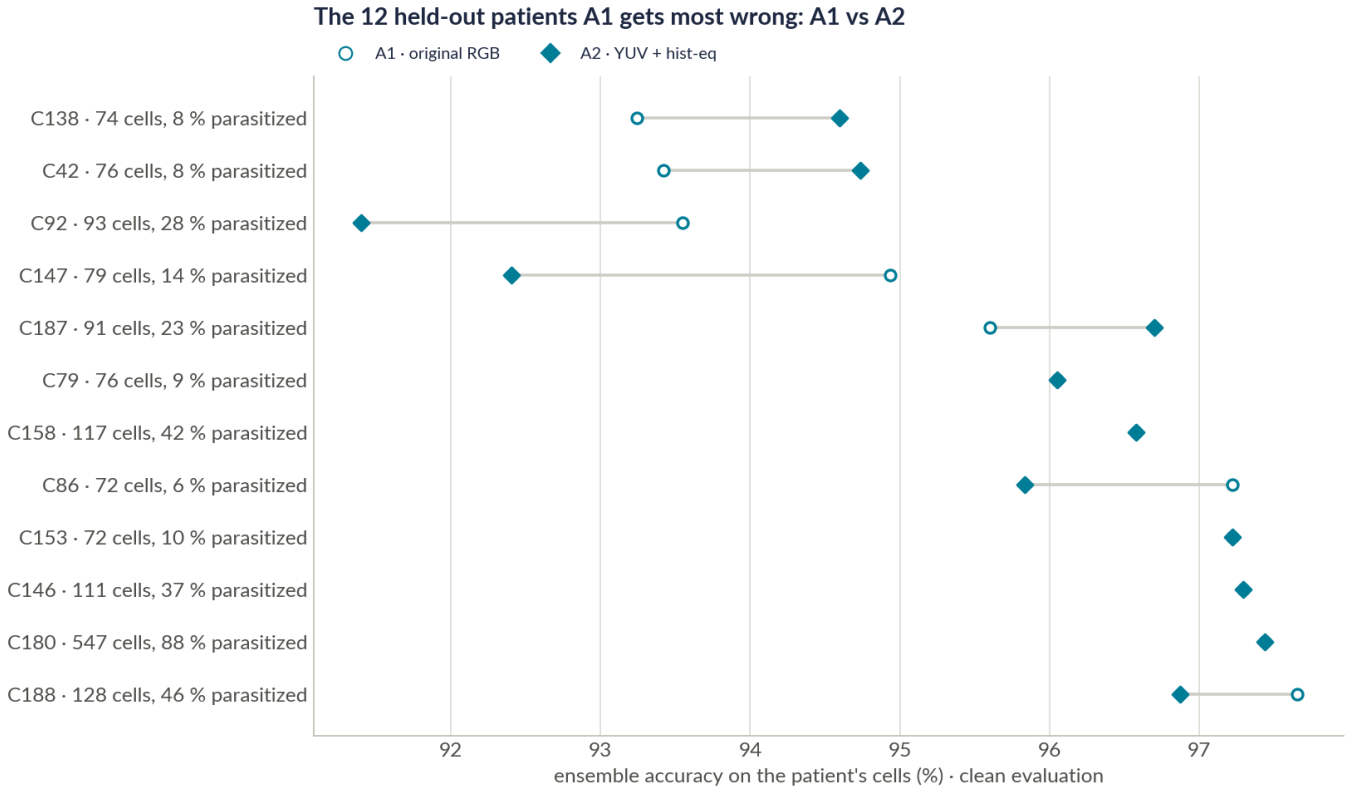
Say it like this. “We were wrong in an informative way. For H1 we got the direction right but the size wrong, because the parasite looks the same in everyone and the augmentation already hides the stain. For H2 we tested a brightness fix, but the stain problem is about colour, and the augmentation already covers brightness.”

Step 6 · Failure analysis

Where the errors are (slide 22)

The A1 ensemble gets **86 of 5,550** unseen-patient cells wrong: **55 missed parasites** (false negatives) and **31 false alarms** (false positives).

- **The errors cluster by patient.** Half of them come from 7 of the 40 patients, and 14 patients have none.
  - A permutation test rejects “errors spread evenly over patients” ($p < 0.001$).
  - The same holds out of fold: 619 errors over 160 patients.
- **False alarms fall on infected smears.**
  - 30 of the 31 are on infected patients’ smears: 1.57 % of their uninfected-labelled cells, against 0.12 % in healthy patients (difference 1.45 pp, 95 % CI 0.82 to 2.14).
  - Out of fold, the rates are 2.35 % and 0.85 %.
  - Two explanations fit. Either the labeller missed some parasites, or these are genuinely ambiguous cells next to parasites (early rings, platelets, stain specks). Without an expert we can’t tell which.
- **Candidate label errors** (confident learning, Northcutt et al. 2021):
  - 8 hold-out cells are flagged (7 labelled parasitized, 1 uninfected), and all 8 are among the errors.
  - Out of fold, 134 cells are flagged.
  - Removing the flagged cells leaves both YUV contrasts unchanged: A2 – A1 stays –0.25 pp, and A3 – A0 moves from 0.00 to +0.04 pp.



Accuracy on each of the 12 held-out patients that A1 gets most wrong (A1 vs A2). A few patients carry most of the errors.

```
def confident_learning_candidates(y, s):
    """A cell is a candidate label error when the other class's probability reaches that
        class's
    average confidence among the cells labelled with it (two-class confident learning)."""
    p_par, p_un = s, 1 - s
    t_par, t_un = p_par[y == 0].mean(), p_un[y == 1].mean()
    ...
```

The blind triage: turning “possible label noise” into our own evidence

- **What the gallery is.** The 86 errors are shuffled with 86 correctly classified cells and numbered 1 to 172, with no labels shown. The pages are `results/phase3b/phase3b_triage_pages.pdf`.
- **What the raters do.** Two teammates each mark every cell: parasite visible (yes / no / unsure), plus one of eight categories.
- **The plan, fixed before anyone rated** (`docs/blind_triage_protocol.md`):
  - Cohen’s κ between the two raters;
  - of the 31 false alarms, how many both raters call “yes”;
  - of the 55 misses, how many both call “no”;
  - the same counts on the correct cells as a baseline;
  - category counts.
- **The scorer.** `src/triage_score.py` implements this plan.
- **The labels are never changed.**
- **Status.** Pending until both sheets are in. If asked: “The protocol and the scoring code are committed. The results come after the two blind ratings.”

Screening view: recall first, no clinical claim (slide 23)

A missed infected cell is costlier than a false alarm, so we also report a recall-first operating point.

- **How it is chosen.** The threshold maximises **F2** (recall weighted 4× over precision) on the pooled out-of-fold **cross-validation** predictions. It is applied **once** to the hold-out set and never tuned on it.

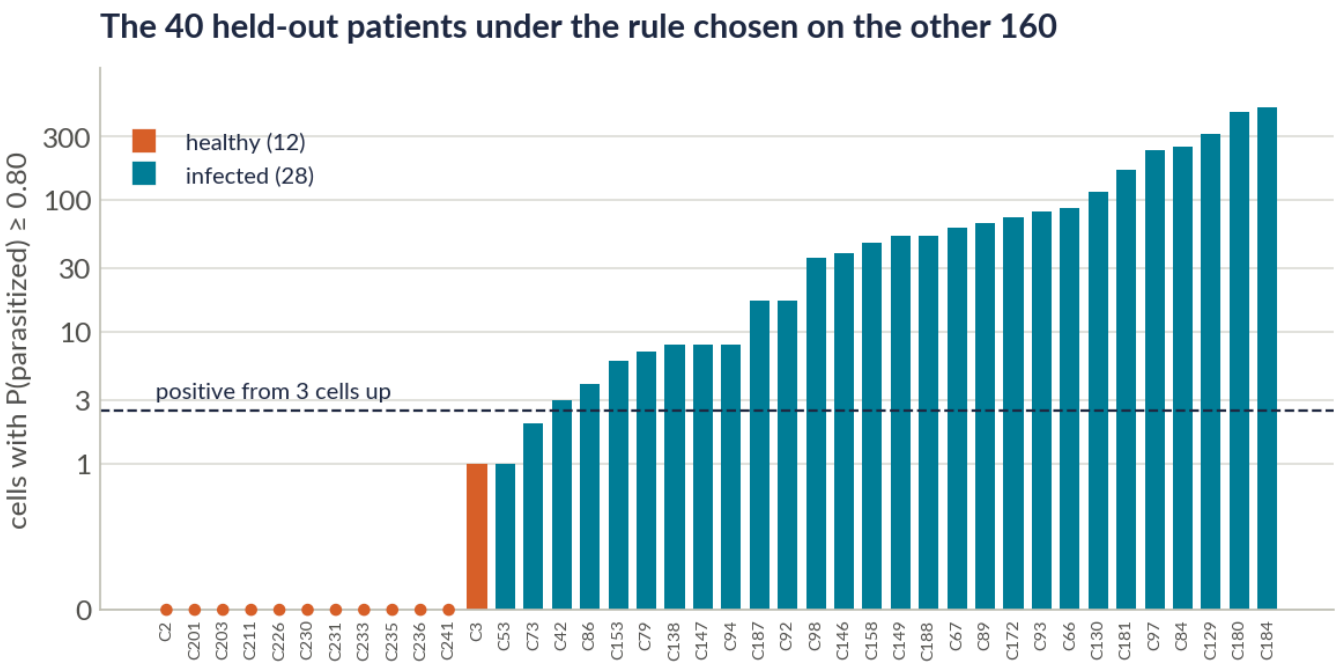
Experiment	Threshold	Recall: at 0.5 → at threshold	Specificity at threshold	ECE
A0 · image split	0.192	97.18 → 98.72 %	95.22 %	0.053
A1 · patient split	0.189	98.05 → **99.22 %**	93.93 %	0.065
A2 · patient split + YUV	0.143	98.22 → 99.22 %	91.74 %	0.060
A3 · image split + YUV	0.159	97.14 → 98.75 %	94.14 %	0.052

- **Calibration.** ECE is 0.05 to 0.07: the probabilities are slightly *under*-confident, as label smoothing (targets of 0.95) predicts.

From cells to patients (slide 24, exploratory)

- **The rule.** A patient is called positive if **at least 3 cells have P(parasitized) ≥ 0.80**.
- **How it was chosen.** Only on the 160 cross-validation patients: the highest Youden J (sensitivity + specificity − 1) among rules with at least 95 % specificity.
  - On those patients: sensitivity 96.7 %, specificity 100 %.

    - With the rule selection nested inside the folds: 96.7 % and 97.4 %.
- **On the 40 unseen patients** (applied once):
    - **26 of 28** infected (92.9 %, exact 95 % CI 76.5 to 99.1);
    - **12 of 12** healthy (100 %, 73.5 to 100).
- **The two misses**, C53 and C73, are the only held-out patients with just three parasitized crops.
- **Why it is exploratory.**
    - NIH kept about 68 uninfected crops per patient but *curated* the parasitized ones.
    - With only one or two parasitized crops per patient, sensitivity falls to 3.6 % and 12.3 %.
    - So these numbers describe aggregation on curated crops, **not diagnostic accuracy on a real smear**.



Patient-level rule on the 40 held-out patients.

Limitations, and how to talk about them

Limitation	What it means
Curated cells, not smears	The patient-level rule is exploratory. NIH chose which crops to keep
One hospital	A patient split removes patient leakage, not site or staining-protocol effects. There is no external test set
One expert's labels	Fuhad et al. set aside 5.1 % of the cells as mislabelled or suspicious. Their list is no longer available, so we keep the original labels, as the paper did
One hold-out draw, one run per fold	A different draw of 40 patients or a second seed would move the numbers. The confidence intervals show by how much
Only 12 healthy held-out patients	The specificity interval is wide (73.5 to 100 %)
Same-cell comparison uses single models	Its interval omits run-to-run noise
Two GPU types (RTX 4090, L40S)	Within a fold, A1, A2 and A3 shared one GPU, so the paired comparisons use identical hardware

No clinical claims.

- **Say:** "screening-oriented analysis", "exploratory", "on curated cells, not a smear", "not supported", "practically zero within ±1 pp".
- **Never say:** "diagnoses malaria", "clinically ready", "proves", "no leakage".

The paper and the code, section by section

The IEEE paper (`Malaria_IEEE_Paper.pdf`, IEEEtran, 8 pages)

Section	What it says	Numbers to know
Title, abstract	*Does 98 % Survive Unseen Patients?* The whole result in 6 sentences	97.62 vs 98.29 %; 98.45 %; 0.85 pp; ±1 pp
I · Introduction	The problem, two questions, five contributions	No new numbers
II · Related Work	11 studies, two gaps	96 to 99.9 %
III · Data	27,558 cells, 200 patients, patient key	10,364 uninfected cells from infected patients
IV-A · Reproduction	The code-exact recipe, 32 tests, the speed-ups	4,223,934 parameters
IV-B · Ablation design	A0 to A3, patient split, YUV	5,550 cells / 40 patients
IV-C · Statistics	Patient bootstrap, Nadeau-Bengio, Holm, decision rules, the exploratory addendum	10,000 resamples
V-A to V-C · Results	Reproduction, H1, leakage on identical cells, H2, equivalence	Tables I to V
V-D · Screening	Recall-first threshold, calibration	98.7 to 99.2 % recall
V-E · Patient level	The rule chosen on CV patients	26/28, 12/12
V-F · Failure analysis	Errors by patient, false alarms, label-error candidates	7 of 40 patients; 1.57 vs 0.12 %
V-G · Compute	11 min per model; 30 models in about an hour	US$6.95
VI · Discussion	Why leakage is small, why one hold-out set isn't enough, why YUV doesn't help	2.3× / 2.6× design effect
VII · Limitations	As in section 9 above	No new numbers
VIII · Conclusion	Four takeaways and four next steps	98.45 %; under 2 pp; ±1 pp; 26/28 and 12/12

The code (`github.com/amathziah/malaria-detection`)

File	What it does
`src/marques2022_exact.py`	The paper-exact recipe: data, splits (image and patient), YUV, model, training, the ensemble (32 tests)
`src/runpod_10fold.py`	Launches one cloud GPU per fold and collects the results
`src/make_ablation_splits.py`	Builds and checks the patient split against NIH's patient mapping; YUV example figure

File	What it does
`src/ablation_stats.py`	Metrics, patient bootstraps, corrected t-tests, Holm, threshold, calibration, verdicts
`src/ablation_figures.py`	The Phase 3 figures
`src/phase3b_analysis.py`	Equivalence, same-cell leakage, the patient-level rule, the error audit, the blinded gallery
`src/triage_kit.py`, `src/triage_score.py`	Triage pages (numbers only) and the pre-planned triage scoring
`docs/ieee/make_tables.py`	Writes every table of the paper from `results/`, so no number is typed by hand
`notebooks/Phase3_Ablation_A1_A3.ipynb`	Trains A1 to A3, one pod per fold
`tests/`	Over 130 automated tests

Trained models. All 40 are public, so nothing has to be retrained:

- release `v1.0-a0-10fold`: the 10 A0 models;
- release `v2.0-ablation`: the 30 A1 to A3 models, with SHA-256 checksums.

Numbers card

Number	Meaning
27,558 / 200	cells / patients (150 infected, 50 healthy); half of the cells parasitized
75 %	of uninfected cells come from infected patients' smears
4,223,934	parameters, exactly the paper's Table 2
97.55 vs 97.56 %	our mean fold accuracy vs the paper's
97.62 vs 98.29 %	our ensemble accuracy vs the paper's (every metric within 1 pp)
99.73 vs 99.76 %	ensemble ROC-AUC, ours vs the paper's
≈ 11 min vs 4 h 46 min	training time per fold, RTX 4090 vs the paper's GTX 1050
5,550 / 40	cells / patients in our patient-level hold-out set (28 infected)
98.45 %	A1 accuracy on 40 unseen patients (95 % CI 97.94 to 98.91); recall 98.05 %, AUC 99.80 %
+1.62 pp	H1's pre-registered test (ensemble MCC; Holm p = 0.15): not supported
−1.65 pp	H1 in cross-validation (fold MCC; Holm p = 0.14): fragile trend; spread 2.5× wider
−0.85 pp	leakage on 17,591 identical cells (MCC; 95 % CI −2.02 to −0.05)
−0.26 pp	H2, YUV on unseen patients (ensemble MCC; 90 % CI −0.73 to +0.16, within ±1 pp): rejected
−0.60 vs 1.51 pp	H2's fold gain vs A1's fold SD: the rejection rule
86 = 55 + 31	A1's errors: missed parasites + false alarms
7 of 40	patients carrying half of the errors
1.57 vs 0.12 %	false-alarm rate on infected vs healthy patients' smears
99.22 %	A1 recall at the threshold chosen on CV (93.93 % specificity)
26/28 · 12/12	infected / healthy unseen patients found by the per-patient rule (exploratory)
0c086ee	the commit that fixed the analysis plan before training

Question drill: 32 likely questions

About the reproduction

1. **Why this paper?** It's a strong result (98.29 %, AUC 99.76 %), and its authors published their executed notebook. That made a code-exact reproduction possible, which almost no other paper allows.
2. **How do you know your rebuild matches?** 32 automated tests compare it with the paper's printed outputs: the parameter count, split sizes, augmentation, image loading, loss and callbacks, and classification reports. The fold models match to 0.01 pp.
3. **Why is your ensemble 0.67 pp lower?** The paper's split was unseeded and can't be recreated, and we use a newer TensorFlow. Our ensemble gains less over its members (0.25 vs 0.60 pp). Every metric is within our 1 pp criterion.
4. **What did the paper get wrong?** Its text swaps precision and recall. Its code scored augmented test images. Its Table 6 "accuracy" is (P + R) / 2. And the text differs from the code in 8 settings.
5. **Did you do feature engineering?** No hand-made features: the CNN learns them. We only resize and use the paper's augmentation. YUV normalisation is the one preprocessing change, and it was our hypothesis.
6. **Why EfficientNet-B0?** It was the paper's model, gives the best accuracy per parameter of its generation, and at 4.2 M parameters is small enough for a phone later.
7. **Why ten folds and an ensemble?** Every learning cell is tested once, we get a mean and a spread, and averaging ten models cancels some individual mistakes.
8. **How did you train 25× faster without changing the results?** We decode once, run the augmentation in parallel with seeds, send uint8 batches to the GPU, compile the step with XLA, and train 10 folds on 10 GPUs. The maths is the same float32.

About the hypotheses

9. **The brief said one change; you made two?** Each experiment changes one thing. Our method change is YUV normalisation (H2). The patient split (H1) changes how we test, not the model; we needed it so H2 is measured on unseen patients. A3 controls for it.
10. **What does "pre-registered" mean, and how can we check?** The hypotheses, primary metric, tests and decision rules were committed to GitHub (commit `0c086ee`, 2 Nov 03:24) before any A1 to A3 model was trained. The later analyses were committed in a dated addendum before they were run. The commit times prove the order.
11. **Why "not supported" for H1 but "rejected" for H2?** H1's rule needed a significant drop to support it, and there wasn't one. H2's rule rejects it if its gain is below the fold spread, and it was (–0.60 vs 1.51 pp).
12. **If leakage exists, why is accuracy on unseen patients higher (98.45 vs 97.64 %)?** Those are different test sets, and the 40-patient draw is easy: the same models score 98.22 % on it but 97.24 % in cross-validation. On *identical* cells, unseen-patient models are slightly worse.
13. **So is there leakage or not?** Yes, but small: 0.85 pp of MCC (about half a point of accuracy) on 17,591 identical cells across 160 patients, and none detectable on the 1,095 cells of the 40 held-out patients.
14. **Why didn't YUV help?** It equalises brightness only, not stain colour. The augmentation already varies brightness and contrast, all images come from one lab, and the equivalence test puts the effect within ±1 pp.
15. **What would have proven you right?** For H1, a significantly lower ensemble MCC on unseen patients. For H2, a fold gain larger than the fold spread. Neither happened.

About the statistics

16. **Why MCC rather than accuracy?** It uses all four confusion-matrix cells and is high only when both classes are right. We fixed it as the primary metric before training. We still report accuracy, recall, specificity and AUC.
17. **Why resample patients rather than cells?** A patient's cells are correlated. Resampling whole patients gives honest intervals: the variance is 2.3× (accuracy) to 2.6× (MCC) what a per-cell bootstrap would claim.
18. **Why did a p-value change from 0.023 to 0.14?** The plan's Welch test treated the ten folds as independent, but any two folds share 8 of 9 training parts. The Nadeau-Bengio correction accounts for that. We report both values and call the result a fragile trend.
19. **What is the Holm correction for?** We test three contrasts, so the chance of one false "significant" rises. Holm adjusts the p-values for that.
20. **Isn't "not significant" enough to say YUV does nothing?** No: a weak test is also "not significant". The equivalence test shows the 90 % CI (–0.73 to +0.16 pp) lies inside ±1 pp, so the effect is practically zero, and the paired design could detect 0.77 pp.
21. **Did you tune anything on the test set?** No. The threshold and the patient rule were chosen on the cross-validation predictions only and applied once to the hold-out set.

About the failure analysis

22. **What kinds of errors does the model make?** 55 missed parasites and 31 false alarms out of 5,550. They cluster in a few patients: half come from 7 of the 40.
23. **Why are almost all false alarms on infected patients' smears?** 30 of 31 are. Either the labeller missed some parasites, or the cells are genuinely ambiguous. We can't tell without an expert, which is why we set up a blind triage.
24. **What is confident learning, and what did it find?** It flags cells whose label disagrees with a confident model. It found 8 hold-out candidates, all among the errors. Removing them changes no verdict.
25. **What is the blind triage?** Two of us rate 172 cells (86 errors and 86 correct) without knowing which is which, using a scoring plan fixed in advance. The labels are never changed.

About screening, patients and the clinic

26. **Why recall first, and how did you pick the threshold?** A missed infection costs more than a false alarm. We chose the threshold with the best F2 on the cross-validation predictions, then applied it once: recall rose to 99.22 % at 93.93 % specificity.
27. **Could this diagnose malaria in a clinic?** Not yet, and we don't claim it. The patient-level result is exploratory, on curated crops from one hospital, with no external test set.
28. **What do 26/28 and 12/12 mean?** A rule chosen only on the 160 CV patients, "positive if at least 3 cells have $P \geq 0.80$", found 26 of 28 infected and all 12 healthy unseen patients. The two misses had only 3 parasitized crops each.

About limits and next steps

29. **What are your main limitations?** Curated cells, one hospital, one labeller, one hold-out draw and one run per fold, and only 12 healthy held-out patients.
30. **What would you do with more time?**
    - Test on another lab's smears (MP-IDB), with a stain-colour augmentation arm.
    - Finish the triage and compare it with an independent relabelling of the dataset.
    - Evaluate patient rules on whole smears.
    - Distil the ensemble into one phone-sized model.

The tough ones

31. **Isn’t your main finding just that the paper was right?** That is a finding. The headline survives a stricter, pre-registered test. We also measured the leakage, found three reporting errors, and showed that a preprocessing gain reported in the literature doesn’t transfer to a strong baseline.
32. **Why trust a 40-patient hold-out set?** We don’t trust it alone. We also cross-validate over 160 patients, pair comparisons on identical cells, and give patient-bootstrap intervals that show the uncertainty.

Glossary of the Phase 3 terms

Term	Plain meaning
Ablation	An experiment that changes one thing to see what it does
Addendum	A dated, committed note fixing extra analyses before running them
Clean / augmented protocol	Scoring the original test images / randomly altered ones, as the paper's code did
Clopper-Pearson interval	An exact confidence interval for a proportion such as 26/28
Cluster (patient) bootstrap	Resampling whole patients, not cells, to get honest intervals
Cohen's κ	Agreement between two raters beyond chance: 0 = chance, 1 = perfect
Confident learning	Flagging cells whose label disagrees with a confident model
Design effect	How much clustering inflates the variance compared with independent cells (2.3× to 2.6× here)
ECE (expected calibration error)	The average gap between predicted probability and observed frequency
Equivalence test (TOST)	Shows an effect is inside a negligible margin (±1 pp), not just "not significant"
F2	The F-score with recall weighted 4× more than precision
Fingerprint	A short hash of a split, checked by every run so all runs use the same split
Holm correction	Adjusts p-values when testing several contrasts
MCC	Matthews correlation coefficient: –1 to 1 (we show × 100), using all four confusion-matrix cells
MDE (minimum detectable effect)	The smallest true effect a design detects 80 % of the time
Nadeau-Bengio correction	Widens the t-test for cross-validation folds that share training data
Out-of-fold (OOF) prediction	A cell's prediction from the fold model that did not train on it
Paired / unpaired comparison	The same cells in both arms / different cells
Patient leakage	Cells of one patient in both training and test, so a test can reward recognising the patient
Patient split	Whole patients go to training or to test, never both
pp (percentage points)	The plain difference between two percentages (98.45 – 97.64 = 0.81 pp)
Pre-registration	Fixing hypotheses, metric, tests and decision rules before the experiment, with a timestamp
Soft voting	Averaging the ten models' probabilities
StratifiedGroupKFold	Folds that keep each group (patient) together and balance the classes

Term	Plain meaning
YUV / luma histogram equalisation	A colour space with brightness (Y) apart from colour (U, V) / stretching Y's histogram over the full range
Youden J	Sensitivity + specificity – 1; used to choose the patient rule